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(1) Research Program/Project/Study Title:	Feasibility Study of a Biomass – Fired Fish Dryer: A Case of Capiz Province
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Mambusao						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								

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(7) Sector/Commodity Agri – Aqua Sector	(8) Discipline/Field of Specialization Business Management
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) The study aimed to determine the viability of producing a biomass-fired fish dryer in the Province of Capiz. The product is a cabinet-type dryer powered by biomass that has a capacity of up to 130 kilograms per batch, considering the size of <i>Manamsi</i> fish (<i>Sardinella Fortunella</i>) and up to 80 kilograms considering the size of boneless and plitted anchovy (<i>dilis</i>). The dryer's size or capacity may be adjusted depending on the needs and budget of interested buyers. The province of Capiz has been increasingly producing dried fish from 3,540 metric tons in 2021 to 5,593 metric tons in 2023. This revealed that the machine is viable regarding the technical, marketing, organization and management, and financial aspects. Resulting in an ROI of 31.20%, IRR of 30.43%, and a payback period of 3.21 years. On the other hand, the firm shall have a capacity of producing 12,377 packs (100g per pack) of boneless dilis and 18,565 packs (200g per pack) in its first-year operations. Considering 5% of the Capiz population with 2.9 kg average consumption, 1% of it will be the patron for the boneless dilis and 4% for the tabagak. The firm will generate an ROI of 34.56% with an Internal Rate of Return of 17.59% and a payback period of 2.49 years.	
(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) dried fish, biomass-fired dried fish, feasibility study, biomass energy, sustainability and agricultural waste utilization	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Dried fish has been considered a vital commodity around the globe. Belton et al (2022) discussed that dried fish products remain a core component of production, trade, diets, and cuisines worldwide. Siddhnath et al (2020) discussed further that the match between dried fish supply and the need for nutrients is vitally suggested by the depleting supply of fisheries resources emphasizing that dry fish can provide high-quality proteins, healthy fats (including long-chain omega-3 fatty acids like eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA)) and a unique source of essential nutrients such as iodine, zinc, copper, selenium, and calcium. Capiz is a province in Western Visayas, Philippines. It's a coastal province near the Sibuyan Sea. Farming and fishing are the primary sources of income for the people in the province. Being dubbed the "Seafood Capital of the Philippines" Capiz is also one of the country's major suppliers of prawn and milkfish. Other agro-industrial harvests include blue marlin, squid, oysters, shrimp, seaweed, squid, and angel wings. Uga or dried fish is among the best delicacies and a major product in the province of Capiz. Belton et al (2022) stated that the dried fish sector employs millions of people, particularly women, who comprise most of the fish-drying workforce in many locations. The same situation is observed in the coastal areas of Capiz where dried fish are processed. The different drying technologies such as solar dryers, charcoal-fired fish dryers, gas-fired fish dryers, and electrical fish dryers have been explored yet there is still a need to	

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consider such costs and the environmental impacts. Engr. Monalyn L. Oloroso, a faculty of the Department of Agricultural and Biosystems Engineering at Capiz State University – Burias Campus has developed “TAPICAP”, a fish dryer with 100 kgs capacity as an aid to produce high-quality dried fish, a cost-efficient and environmentally friendly technology. A market study was conducted by Nexus Agribusiness Solutions to determine the marketability of the machine and so the viability of the same technology, thus this study.

The study was conducted to determine the feasibility of the biomass–fired fish dryer in terms of the different aspects such as technical; marketing; organization and management; financial and socio – economic aspects in the perspective of the following:

1. biomass – fired fish dryer fabricator and
2. dried fish processors.

THEORETICAL AND CONCEPTUAL FRAMEWORK

The study was anchored on Bowen et al (2009) feasibility studies are used to determine whether an intervention is appropriate for further testing; in other words, they enable researchers to assess whether or not the ideas and findings can be shaped to be relevant and sustainable. Such research may identify not only what if anything in the research methods or protocols needs modification but also how changes might occur.

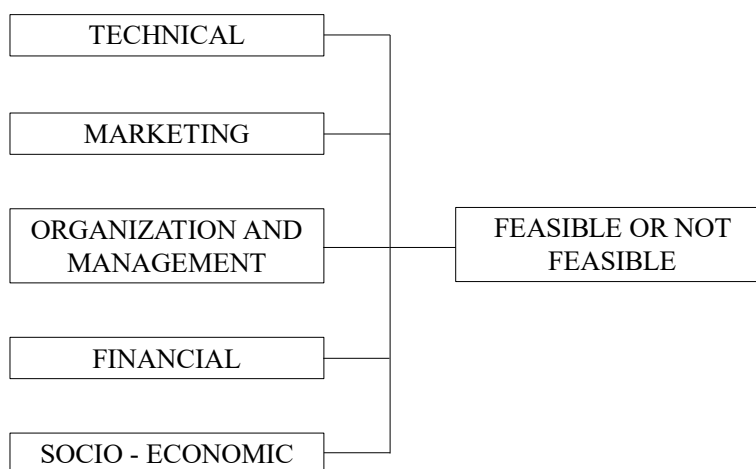


Figure 1. Schematic diagram of the study.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted through document analysis, actual observations of the machine operations, and random interviews. The results of the market study conducted by Manalili et al (2024) were considered in the market analysis. Random interviews among fabricators and dried fish processors were also conducted. Descriptive analysis and financial analysis were used.

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(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

BIOMASS – FIRED FISH DRYER FABRICATOR

Technical Aspect

Product

The product (TAPITAP) is a cabinet-type biomass-fired fish dryer with an average drying capacity of 100 kg. The machine has passed the AMTEC Test conducted last February 5, 2024. It is a fabricated machine made from food-grade materials thus, it is hygienic to use and can produce quality dried fish. Utilizing the fabrication process, the fish dryer can be fabricated by a fabricator and a helper in one month. The total materials cost for each machine is Php338,250.00. In the first year of operation, it is assumed that 10 units of TAPITAP will be produced and sold, and it is projected to increase by 1 unit yearly.



Image credit: Engr. Monalyn Oloroso

Capacity, Volume, and Cost of Production

The proposed firm will produce 10 machines in a year. The cost of materials to create a machine is estimated at Php338,250.00 (based on prototype production). The firm will hire 1 fabricator and an assistant fabricator.

Fabrication Materials

The materials needed shall be outsourced from the firm's trusted suppliers. Each material will undergo a quality check to ensure quality and efficient machine will be produced. This is to guarantee that food-grade dried fish is produced out of TAPICAP.

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Marketing Aspect

Target Market and Projected Demand

The target market for the biomass–fired fish dryer will be the dried fish processors in the province of Capiz. The result of the market study conducted by Manalili et al (2024) showed an increase in dried fish production from 3,540 metric tons in 2021 to 5,593 metric tons in 2023. Moreover, the expanding list of destinations for dried fish implies a widening market reached is probably driven by increasing demand by local and distant regions. The increase in trade value against fluctuating volumes suggests that pricing dynamics and value addition in the trade of dried fish products may also be critical factors contributing to the industry's growth. This shall give a significant market for the product as shown in Table 1 if all the dried fish producers opt for mechanized operations.

In consideration of the cost of the machine and the living conditions of the majority of the dried fish processors in the province, it is projected that the demand for the biomass–fired fish dryer will be 5% of the units of machine needed to produce dried fish as based on the total projected production capacity of the province.

Table 1. Units of machine needed based on the volume of dried fish produced.

Year	Volume of Production (in tons)	Units of Machine (based on 100 kgs capacity)
2025	7,645	2,312
2026	8,671	2,662
2027	9,697	2,932
2028	10,723	3,243
2029	11,749	3,553

Table 2. Projected demand for biomass-fired fish dryers.

Year	Units of Machine	Projected Demand
2025	2,312	116
2026	2,622	131
2027	2,932	147
2028	3,243	162
2029	3,553	178

When considering the purchase of a fish dryer in the Philippines, it is important to evaluate the brands available in the market. Comparing prices, features, and specifications from various manufacturers can help decision-making. Assessing the long–term benefits, energy efficiency, environmental effects and overall value for money is crucial in determining the most suitable fish dryer machine. Being accessible to the fish dryer processors is an advantage for the firm.

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The firm can offer a lifetime service warranty for the machine purchased by the customers. Through this, the firm will be able to maintain good relationships with the customers and can encourage other processors to consider buying and using the machine.

Competitors

Identified competitors of the TAPICAP are from abroad and from Mindanao. This will imply additional costs for buying the machine due to higher delivery costs. It was also noted that most of the existing fish dryers in the market are fabricated not only for dried fish production but also for fruits and vegetables, thus, drying capacity may vary.

Competitor 1.



Product Type: Continuous multi-layer mesh belt dryer
Output: 200 – 3,000 kg/hr
Drying Time: 0.5h – 5h
Temperature: 40 – 120°C
Power Source: 23 kw to 150 kw
Heat source: coal, wood chips, natural gas, diesel, propane, electricity, etc.
Customization: can be customized
Applicable: vegetable/fruits/herbs/ agriculture etc.
Core Advantages: continuous automatic work/large output/dryer time short/flexible heat source selection

Source: <https://www.bxdrymachine.com/fish-dryer.html>



Power supply: 220V/50 Hz/ 60Hz
Power input: 1.01 kw
Net weight: 150 kg
Dimension: 1180:690:1800mm
Lead Time: 10 – 30 days
Warranty: 2 years
Capacity: 20 – 100 kgs

Source: <https://www.ike.cn/wrh-100t-research-type-commercial-food-dehydrator>

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shuliy fish dryer machines

Source: <https://dryingplant.com/fish-dryer-machines-price-philippines/>



Source: Facebook Page of Sec. Manny Piñol (November 13, 2020)

Image of the fish drying facility developed by Engr. Rudy Cane of Agusan del Norte costing about Php300,000.00.

Projected Sales

The firm will produce 10 units in its first year of operation costing Php500,000.00 for each machine. The projected sales of the firm is shown in Table 3 below.

Table 3. Projected Sales for the next five years.

Year	Unit Sales	Unit Price	Total Sales
2025	10	500,000.00	5,000,000.00
2026	11	550,000.00	6,050,000.00
2027	12	605,000.00	7,260,000.00
2028	13	665,500.00	8,651,500.00
2029	14	732,050.00	10,248,700.00

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Organization and Management

Sole proprietorship will be the form of business organization and shall have two (2) employees, the fabricator, and the assistant fabricator. The fabricator shall also serve as the manager of the firm at the same time in charge of the production. The manager shall receive a salary of Php35,000.00 per month and the asst. fabricator shall have a monthly salary of Php20,000.00. Both will be in-charge in the overall wellness of the production area and office. They will be tasked also to deliver the machine.

Financial

. The business will need an initial investment of Php1,060,000.00. Considering the projected five-year operations, the proposed firm shall have an average income for the next five years of Php330,255.29. The different assumptions and plans for fish–dryer production resulted in a net present value of Php1,243,424.00. This indicates that the proposed business will gain and generate higher cash inflows than cash outflows over the next five years. Moreover, it will generate a Return on investment of 31.20%, an Internal Rate of Return of 30.43%, and a payback period of 3.21 years. The favorable results connote possible good performances of the proposed business.

Socio-Economic Aspects

The production of biomass–fired dryers will entail additional income to the economy through employment opportunities and taxes for the government. The proposed business will hire 2 persons as fabricators and an assistant fabricator of the machine and was expected to increase based on the increase of the machine demand. Moreover, it will be paying an average amount of taxes for the next five years amounting to Php155,414.40.

DRIED FISH PROCESSORS

Technical Aspect

Product

The proposed products are boneless dilis and tabagak. They will be processed using the technique known among dried–fish processors in Capiz for quality and hygienic considerations a biomass-fired fish dryer will be used.

Production Materials

The materials needed shall be outsourced from the trusted fish suppliers of the firm. The raw materials will undergo a quality check before they will be able to reach the production area. The firm will need a consistent supply of raw materials in order to sustain its production.

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Projected Production

Table 1. Production capacity of the firm (boneless dilis).

Year	Projected Production	Production in kg	Production in packs (100g)
2025	123,769	1,238	12,377
2026	125,229	1,252	12,523
2027	126,707	1,267	12,671
2028	128,202	1,282	12,820
2029	129,715	1,297	12,971

Table 2. Production capacity of the firm (tabagak).

Year	Projected Production	Production in kg	Production in packs (200g)
2025	123,769	4,951	24,754
2026	125,229	5,009	25,046
2027	126,707	5,068	25,341
2028	128,202	5,128	25,640
2029	129,715	5,189	25,943

Marketing Aspect

Target Market and Projected Demand

The target market for the biomass-fired fish dryer will be the dried fish processors in the province of Capiz. The result of the market study conducted by Manalili et al (2024) showed an increase in dried fish production from 3,540 metric tons in 2021 to 5,593 metric tons in 2023. Moreover, the expanding list of destinations for dried fish implies a widening market reached is probably driven by increasing demand by local and distant regions. The increase in trade value against fluctuating volumes suggests that pricing dynamics and value addition in the trade of dried fish products may also be critical factors contributing to the industry's growth. Considering the increase in the province's population, which is 1.18%, and the country's average consumption of dried fish, Table 1 shows the demand for dried fish for the next five years.

Table 1. Projected demand for dried fish.

Year	Projected Population	Projected Demand
2025	853,578	2,475,377
2026	863,651	2,504,586
2027	873,842	2,534,141
2028	884,153	2,564,043
2029	894,586	2,594,299

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Projected Sales

The firm will produce 12,377 packs (100g per pack) of boneless dilis and 24,754 packs (200g per pack) of tabagak for its first year of operations. It is projected to increase based on the projected increase in population. The projected sales of the firm is shown in Table 3 below.

Table 3. Projected Sales for the next five years.

Year	Projected Sales		
	Boneless Dilis	Tabagak	Total
2025	1,704,916	2,939,510	4,644,426
2026	1,795,488	3,095,669	4,891,157
2027	1,890,666	3,259,770	5,150,436
2028	1,990,679	3,432,204	5,422,883
2029	2,095,763	3,613,384	5,709,147

Organization and Management

A sole proprietorship will be the form of business organization and shall have two (2) permanent employees, the manager, and the machine operator. The manager shall be in charge of supervising the whole operations of the business including production and marketing. The manager shall receive a salary of Php25,000.00 per month and the machine operator shall have a monthly salary of Php20,000.00. Both will be in-charge in the overall wellness of the production area and office. They will also be tasked to deliver the products. On-call personnel who will be in charge of deboning (Php300.00/banyera) and laying of fish in trays (Php5.00/tray) shall also be employed.

Financial

The business will need an initial investment of Php1,074,000.00. Considering the projected five-year operations, the proposed firm shall have an average income for the next five years of Php371,199.61. The different assumptions and plans for fish-dryer production resulted in a net present value of Php488,091.00. This indicates that the proposed business will gain and generate considerable cash inflows for the next five years. Moreover, it will have a Return on investment of 34.56%, an Internal Rate of Return of 17.59%, and a payback period of 2.49 years. The favorable results connote possible good performances of the proposed business.

Socio-Economic Aspects

The production of biomass-fired dryers will entail additional income to the economy through employment opportunities and taxes for the government. It will help reduce waste using rice hull as its energy source. Moreover, it will be paying an average amount of taxes for the next five years amounting to Php174,682.17.

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Based on the foregoing findings, the following conclusions were drawn.

1. The production and marketing of biomass-fired fish dryers is feasible in terms of technical, marketing, organization and management, financial, and socio-economic aspects but still needs enhancement to further enhance the profitability of the project.
2. The production and marketing of dried fish using biomass-fired fish dryer is feasible in terms of technical, marketing, organization and management, financial, and socio-economic aspects but has to see to it the consistent supply of raw materials in order to stabilize costs and enhance profitability.

Recommendations

- Looking into suppliers of food-grade materials used in the production of biomass-fired fish dryers to minimize costs.
- Disseminate the importance of preparing foods in compliance with standards.
- Due to the decline of raw materials (tabagak), strict implementation of closed fishing season (Bureau Administrative Circular No. 255) to conserve and protect economically important species during their spawning period.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Technology Commercialization Plan

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Production of the machine shall have an estimated ROI of 31.20%, IRR of 30.43%, and a payback period of 3.21 years. While production of dried fish using the machine will result to an ROI of 34.56% with an Internal Rate of Return of 17.59% and a payback period of 2.49 years.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The researchers would like to extend gratitude to the CAPSU Management and the Department of Science and Technology – Technology Application and Promotion Institute (DOST) which funded the study entitled The Enhancement, Pilot Testing, and Market Study of a Biomass-Fired Fish Dryer under the Innovation for Commercialization (TECHNiCOM) Program.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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